

Paragliding Intelligence Platform

Volume 04 - API Specification v2 Technical

Implementation-ready REST, WebSocket, event, authentication, entitlement, and error-contract specification

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Scope: This document replaces outline-level API notes with concrete endpoint contracts for the paragliding-first meteo platform. It assumes the PRD v2, Technical SAD v2, and Database Design v2 are the upstream sources of truth.

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1. API Design Principles

The API is designed to be product-safe, mobile-friendly, GIS-first, and weather-model-aware. The platform is not a generic weather API; it is a decision-intelligence system whose main output is whether a specific pilot should drive to, launch from, continue flying from, or avoid a paragliding site at a given time.

Every decision endpoint must preserve traceability. A GO, MAYBE, or NO-GO output must reference the forecast model run, launch-site configuration, safety blockers, skill profile, and scoring version used to generate it. This prevents invisible changes in model data from producing unexplained recommendations.

Every endpoint that returns aviation- or safety-relevant information must separate measured facts, forecast-derived values, corrected predictions, and AI-generated summaries. The client UI must never show LLM text as if it were raw meteorological data.

The mobile app must continue functioning in weak connectivity. APIs therefore support compact payloads, differential sync, ETags, offline map packages, and WebSocket reconnect cursors.

Principle	Implementation rule
Contract-first	All services expose OpenAPI 3.1 specs and JSON Schema components. Breaking changes require /v2 or feature flags.
Safety traceability	Decision responses include run_id, scoring_version, blocker list, uncertainty, and data_age_seconds.
Tier-aware	Entitlements are checked at API Gateway and again in owner services for high-value endpoints.
Cacheable reads	Static launch data, airspace, and map metadata use ETag and stale-while-revalidate.
Realtime by WebSocket	Tracking, live alerts, station feeds, and event operations use WebSocket channels plus resumable cursors.
No hallucinated weather	AI endpoints must cite structured context IDs and refuse unsupported claims.

2. Public API Domains and Service Ownership

Domain	Base path	Owner service	Primary storage	Realtime channel
Identity	/v1/auth, /v1/users	auth-service	PostgreSQL users schema	none
Pilot profile	/v1/pilot-profiles	profile-service	PostgreSQL user/profile tables	profile.updated
Launch sites	/v1/sites	site-service	PostGIS site schema	sites.changed
Weather	/v1/weather	forecast-api-service	TimescaleDB + object storage	weather.model_run.compl eted
Flyability	/v1/flyability	decision-service	TimescaleDB score tables	flyability.updated
Risk	/v1/risk	risk-service	TimescaleDB risk tables	risk.alert
AI briefing	/v1/briefings	ai-briefing-service	PostgreSQL ai_context + ai_briefings	briefing.ready
XC planner	/v1/xc	xc-planner-service	PostGIS route tables	xc.plan.ready
Tracking	/v1/tracks, /ws/tracking	tracking-service	TimescaleDB telemetry	track.*
Alerts	/v1/alerts	notification-service	PostgreSQL alerts + event log	notification.sent
Community	/v1/observations	observation-service	TimescaleDB observation tables	observation.created
Digital twins	/v1/twins	digital-twin-service	PostgreSQL + TimescaleDB features	twin.model.updated
Airspace	/v1/airspace	airspace-service	PostGIS airspace schema	airspace.changed
Billing	/v1/billing, /v1/entitlements	billing-service	PostgreSQL subscription tables	entitlement.changed
Admin/Ops	/v1/admin, /v1/ops	ops-service	PostgreSQL audit + metrics bridge	ops.incident

3. Authentication, Authorization, and Sessions

The API uses OAuth2-compatible passwordless/email or social login for consumers, organization-managed invitations for clubs/schools, and scoped service tokens for internal services. Access tokens are short-lived JWTs. Refresh tokens are opaque, stored hashed, and rotated on use.

Role	Capabilities
anonymous	Read public launch summaries, public map metadata, public safety notices.
free_pilot	Manage profile, favorites, basic forecasts, limited alerts, community reports.
pro_pilot	Advanced forecast windows, full flyability timeline, unlimited favorites, configurable alerts.
pro_plus_pilot	Model comparison, AI briefing, XC planner, corrected/digital-twin outputs.
club_admin	Manage club sites, members, station integrations, club dashboards.
school_instructor	Student-safe windows, training sessions, instructor dashboard.
event_admin	Competition task planning, live tracking, event weather center.
rescue_operator	Emergency track lookup, drift/search-area tools, audit-heavy access.
platform_admin	System operations, moderation, data-quality overrides.

POST /v1/auth/login/email-link

Field	Value
Owner service	auth-service
Minimum tier	Free
Summary	Start passwordless login by email.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "email": "pilot@example.com",
  "locale": "en-NO",
  "redirect_uri": "paraflly://auth/callback"
}
```

Response schema / example

```
{
  "status": "email_sent",
  "expires_in_seconds": 900
}
```

HTTP	Code	Condition
400	invalid_email	Email is malformed.
429	rate_limited	Too many login attempts.

POST /v1/auth/token/exchange

Field	Value
Owner service	auth-service
Minimum tier	Free
Summary	Exchange signed login code for access and refresh token.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "login_code": "opaque-code-from-email",
  "device_id": "ios:8f2...",
  "device_name": "iPhone 15 Pro"
}
```

Response schema / example

```
{
  "access_token": "jwt",
  "refresh_token": "opaque_refresh_token",
  "expires_in": 900,
  "user": {"id": "usr_...", "email": "pilot@example.com"}
}
```

HTTP	Code	Condition
400	invalid_code	Login code is invalid or expired.
409	device_blocked	Device was administratively blocked.

POST /v1/auth/refresh

Field	Value
Owner service	auth-service
Minimum tier	Free
Summary	Rotate refresh token and issue new access token.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "refresh_token": "opaque_refresh_token",
  "device_id": "ios:8f2..."
}
```

Response schema / example

```
{
  "access_token": "jwt",
  "refresh_token": "new_opaque_refresh_token",
  "expires_in": 900
}
```

HTTP	Code	Condition
401	invalid_refresh	Refresh token invalid, revoked, or expired.

GET /v1/entitlements/me

Field	Value
Owner service	billing-service
Minimum tier	Free
Summary	Return effective tier and feature flags for the authenticated user.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=60

Response schema / example

```
{
  "user_id": "usr_123",
  "tier": "pro_plus",
  "features": {
    "ai_briefing": true,
    "xc_planner": true,
    "model_comparison": true,
    "max_alert_rules": 50,
    "forecast_horizon_hours": 240
  },
  "source": "stripe_subscription",
  "valid_until": "2026-07-04T00:00:00Z"
}
```

}

HTTP	Code	Condition
401	unauthorized	Missing or invalid token.

4. Common Headers, Envelope, Pagination, Filtering, and Idempotency

Header	Required	Description
Authorization: Bearer <jwt>	most endpoints	JWT access token. Anonymous endpoints explicitly state otherwise.
X-Client-Version	yes	Semantic app version used for compatibility, kill switches, and analytics.
X-Request-ID	yes	Client-generated UUID used across logs/traces.
Idempotency-Key	write endpoints	Required for POST/PATCH/PUT that create or mutate state.
If-None-Match	cacheable GET	Used with ETag for launch/site/airspace metadata sync.
X-Timezone	optional	IANA timezone for human-friendly response labels; UTC remains canonical.
Accept-Encoding: br,gzip	recommended	Mobile bandwidth optimization.

Standard success envelope

```
{
  "data": {},
  "meta": {
    "request_id": "req_...",
    "api_version": "v1",
    "server_time": "2026-06-04T10:15:00Z",
    "data_age_seconds": 420
  }
}
```

Standard list pagination uses cursor pagination for realtime-friendly data and offset pagination only for admin back-office tables.

```
{
  "data": [{"id": "site_123"}],
  "page": {
    "next_cursor": "eyJvZmZzZXQiOjEwMH0=",
    "has_more": true,
    "limit": 50
  }
}
```

Filtering grammar

Pattern	Example	Rule
bbox	?bbox=minLon,minLat,maxLon,maxLat	Spatial queries must specify WGS84 lon/lat.
time range	?from=2026-06-05T00:00:00Z&to=2026-06-05T18:00:00Z	UTC timestamps only.
include	?include=hazards,landings,latest_score	Embeds related objects only from explicit allow-list.
fields	?fields=id,name,score,status	Sparse fieldsets for mobile payload reduction.
skill profile	?pilot_level=beginner	Overrides authenticated profile only for what-if calculations.

5. Error Model and Observability Contract

```
{
  "error": {
    "code": "forecast_unavailable",
    "message": "No forecast cube is available for the requested site and time range.",
    "details": {
      "site_id": "site_123",

```

```

    "requested_from": "2026-06-05T10:00:00Z",
    "required_model": "ecmwf_hres"
  },
  "retry_after_seconds": 600,
  "trace_id": "trc_01HX..."
}
}

```

HTTP	Class	Examples
400	client validation	invalid_bbox, invalid_time_range, invalid_profile_input
401	authentication	unauthorized, token_expired
403	authorization/entitlement	tier_required, role_required, organization_access_denied
404	not found	site_not_found, track_not_found
409	conflict	duplicate_report, stale_revision, route_no_longer_valid
422	domain impossible	launch_direction_incompatible, invalid_task_geometry
429	rate limit	rate_limited, websocket_connection_limit
503	dependency failure	forecast_unavailable, station_feed_unavailable, ai_provider_unavailable

Every service must log request_id, trace_id, user_id hash, organization_id, endpoint, status, duration_ms, cache_result, entitlement_decision, and service_dependency_summary. Decision endpoints must also log scoring_version, model_run_ids, and blocker codes.

6. User, Pilot Profile, and Device APIs

GET /v1/users/me

Field	Value
Owner service	profile-service
Minimum tier	Free
Summary	Read own account and profile summary.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=60

Response schema / example

```
{
  "id": "usr_123",
  "display_name": "Vladimir",
  "email": "pilot@example.com",
  "locale": "en-NO",
  "units": {"wind": "kmh", "altitude": "m"},
  "pilot_profile_id": "prof_456",
  "tier": "pro_plus"
}
```

HTTP	Code	Condition
401	unauthorized	Missing token.

PUT /v1/pilot-profiles/me

Field	Value
Owner service	profile-service
Minimum tier	Free
Summary	Create or replace pilot risk profile.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "pilot_level": "intermediate",
  "wing_class": "EN-B",
  "flight_types": ["hike_and_fly", "xc"],
  "max_surface_wind_kmh": 24,
  "max_gust_kmh": 32,
  "risk_tolerance": "normal",
  "home_location": {"lat": 59.9139, "lon": 10.7522}
}
```

Response schema / example

```
{
  "id": "prof_456",
  "scoring_profile_version": "pilot-profile-v3",
  "updated_at": "2026-06-04T10:15:00Z"
}
```

HTTP	Code	Condition
400	invalid_profile_input	Limits outside safe supported ranges.

POST /v1/devices

Field	Value
Owner service	profile-service
Minimum tier	Free

Summary	Register mobile device for push notifications and emergency metadata.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "platform": "ios",
  "push_token": "apns-token",
  "device_name": "iPhone 15",
  "app_version": "1.0.0"
}
```

Response schema / example

```
{"device_id": "dev_123", "push_enabled": true}
```

HTTP	Code	Condition
400	invalid_push_token	Push token is rejected by validation.

7. Launch Site, Landing, Hazard, and Site-Intelligence APIs

Launch-site APIs are GIS-first. The mobile map primarily uses bbox-based sync, while site detail screens use include parameters to hydrate landings, hazards, local rules, webcams, stations, and latest flyability score.

GET /v1/sites

Field	Value
Owner service	site-service
Minimum tier	Anonymous/Free
Summary	List launch sites in bounding box or near a coordinate.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=300, stale-while-revalidate=3600

Response schema / example

```
{
  "data": [
    {
      "id": "site_vaga_001",
      "name": "Vaga Launch",
      "country": "NO",
      "location": {"lat": 61.875, "lon": 9.096},
      "launch_altitude_m": 950,
      "allowed_wind_sectors": [{"from_deg": 285, "to_deg": 45}],
      "difficulty": "intermediate",
      "latest_status": "GO"
    }
  ],
  "page": {"next_cursor": null, "has_more": false}
}
```

HTTP	Code	Condition
400	invalid_bbox	Bounding box missing or too large.

GET /v1/sites/{site_id}

Field	Value
Owner service	site-service
Minimum tier	Free
Summary	Read full launch-site record.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=600

Response schema / example

```
{
  "id": "site_vaga_001",
  "name": "Vaga Launch",
  "geometry": {"type": "Point", "coordinates": [9.096, 61.875]},
  "launch_altitude_m": 950,
  "landings": [{"id": "landing_01", "altitude_m": 370}],
  "hazards": [
    {"type": "rotor_zone", "severity": "high", "geometry_id": "haz_998"}
  ],
  "local_rules": ["Respect landowner access road", "Avoid landing field during livestock season"],
  "weather_station_ids": ["station_77"],
  "webcam_ids": ["cam_44"]
}
```

HTTP	Code	Condition
404	site_not_found	Unknown site id.

ETag required for offline sync.

POST /v1/sites/{site_id}/revision-proposals

Field	Value
Owner service	site-service
Minimum tier	Free
Summary	Submit proposed edit to launch, landing, hazard, or local rule.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "proposal_type": "hazard_add",
  "summary": "New power line near lower landing",
  "geometry": {"type": "LineString", "coordinates": [[9.09,61.87],[9.10,61.88]]},
  "evidence": [{"type": "photo", "attachment_id": "att_123"}]
}
```

Response schema / example

```
{"proposal_id": "prop_123", "status": "pending_review"}
```

HTTP	Code	Condition
403	role_required	User must be verified or trusted contributor for direct edits.

8. Weather Forecast, Map Tile, Model-Run, and Grid-Query APIs

Weather endpoints expose normalized platform data, not raw upstream provider payloads. Model provenance is preserved through `model_run_id` and `provider_run_time`. Mobile clients should prefer site forecast endpoints for decision screens and vector/raster tile endpoints for maps.

GET /v1/weather/model-runs

Field	Value
Owner service	forecast-api-service
Minimum tier	Free
Summary	Return available forecast model runs by provider, region, and time.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=300

Response schema / example

```
{
  "data": [
    {
      "model_run_id": "run_ecmwf_20260604_00",
      "provider": "ecmwf_open_data",
      "model": "ifs",
      "run_time": "2026-06-04T00:00:00Z",
      "available_until": "2026-06-14T00:00:00Z",
      "status": "ready",
      "variables": ["wind_u", "wind_v", "gust", "temperature", "precipitation"]
    }
  ]
}
```

HTTP	Code	Condition
503	forecast_unavailable	No current model run ready.

GET /v1/weather/sites/{site_id}/hourly

Field	Value
Owner service	forecast-api-service
Minimum tier	Free/Pro
Summary	Return normalized hourly site forecast. Free tier may receive shorter horizon and fewer variables.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private/public mixed; ETag by site_id+model_blend_id+horizon

Response schema / example

```
{
  "site_id": "site_vaga_001",
  "model_blend_id": "blend_nordic_v2_20260604_06",
  "hours": [
    {
      "time": "2026-06-05T12:00:00Z",
      "surface_wind_kmh": 14.2,
      "surface_wind_dir_deg": 315,
      "gust_kmh": 23.0,
      "wind_1000m_kmh": 21.4,
      "cloudbase_m_agl": 1350,
      "thermal_strength_ms": 2.1,
      "precip_probability": 0.05,
      "thunderstorm_probability": 0.01,
    }
  ]
}
```

```

    "model_confidence": 0.82
  }
]
}

```

HTTP	Code	Condition
403	tier_required	Requested variables or horizon require higher tier.
404	site_not_found	Unknown site id.

GET /v1/weather/grid/point

Field	Value
Owner service	forecast-api-service
Minimum tier	Pro+
Summary	Query gridded normalized weather at any coordinate and altitude.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=180 for same coordinate bucket

Response schema / example

```

{
  "lat": 61.875,
  "lon": 9.096,
  "altitude_m_msl": 1400,
  "time": "2026-06-05T12:00:00Z",
  "values": {
    "wind_kmh": 22.8,
    "wind_dir_deg": 310,
    "temperature_c": 11.4,
    "relative_humidity": 0.61,
    "vertical_velocity_ms": 0.4
  },
  "source": {"model_run_ids": ["run_ecmwf_20260604_00", "run_metno_20260604_06"]}
}

```

HTTP	Code	Condition
403	tier_required	Grid point API requires Pro+.
422	altitude_out_of_range	Altitude outside model domain.

GET /v1/weather/tiles/{layer}/{z}/{x}/{y}

Field	Value
Owner service	tile-service
Minimum tier	Free/Pro
Summary	Return raster or vector weather tile for map rendering.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	CDN cached by layer/time/model/tile

Response schema / example

Binary response or MVT/PNG depending on layer. Headers:

Content-Type: application/x-protobuf | image/png

ETag: tile-layer-time-model-z-x-y

Cache-Control: public, max-age=600

HTTP	Code	Condition
404	tile_not_found	Tile unavailable for time/model.

GET /v1/weather/animation/{layer}/manifest

Field	Value
Owner service	tile-service
Minimum tier	Free/Pro
Summary	Return animation frame manifest for timeline slider.

Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=300

Response schema / example

```
{
  "layer": "wind_surface",
  "frames": [
    {"time": "2026-06-05T12:00:00Z", "tile_template": "/v1/weather/tiles/wind_surface/{z}/{x}/{y}?t=2026060512"}
  ],
  "legend": {"unit": "kmh", "min": 0, "max": 60}
}
```

HTTP	Code	Condition
400	invalid_layer	Layer not supported for current tier.

9. Flyability, Risk, Thermal, Cloudbase, and Decision APIs

Decision APIs are the core differentiator. They produce understandable and traceable output, not just raw meteorological values. The hard-blocker list always overrides numerical score. Example hard blockers: thunderstorm probability over threshold, crosswind outside site envelope, gust over profile limit, cloudbase below minimum, active weather warning, or official site closure.

GET /v1/flyability/sites/{site_id}/timeline

Field	Value
Owner service	decision-service
Minimum tier	Free/Pro
Summary	Return hourly GO/MAYBE/NO-GO timeline for a site.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=300

Response schema / example

```
{
  "site_id": "site_vaga_001",
  "scoring_version": "flyability-v4.2",
  "pilot_profile_applied": "prof_456",
  "hours": [
    {
      "time": "2026-06-05T12:00:00Z",
      "status": "GO",
      "flyability_score": 83,
      "safety_score": 78,
      "xc_score": 71,
      "confidence": 0.84,
      "blockers": [],
      "reasons": [
        {"code": "wind_aligned", "text": "Wind direction matches launch sector."},
        {"code": "gust_margin_ok", "text": "Gusts remain below profile limit."}
      ],
      "source": {"model_blend_id": "blend_nordic_v2_20260604_06"}
    }
  ]
}
```

HTTP	Code	Condition
403	tier_required	Requested horizon/features require Pro.
503	forecast_unavailable	No forecast blend ready.

POST /v1/flyability/what-if

Field	Value
Owner service	decision-service
Minimum tier	Pro
Summary	Evaluate temporary pilot/site/time assumptions without changing profile.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_id": "site_vaga_001",
  "from": "2026-06-05T10:00:00Z",
  "to": "2026-06-05T18:00:00Z",
  "pilot_assumptions": {
```



```

    "pilot_level": "beginner",
    "wing_class": "EN-A",
    "max_surface_wind_kmh": 20,
    "max_gust_kmh": 26
  }
}

```

Response schema / example

```

{
  "best_window": {"from": "2026-06-05T11:00:00Z", "to": "2026-06-05T13:30:00Z"},
  "overall_status": "MAYBE",
  "hours": []
}

```

HTTP	Code	Condition
400	invalid_profile_input	What-if limits invalid.

GET /v1/risk/sites/{site_id}/timeline

Field	Value
Owner service	risk-service
Minimum tier	Pro/Pro+
Summary	Return detailed risk vector for a site.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=300

Response schema / example

```

{
  "site_id": "site_vaga_001",
  "risk_model_version": "risk-v3.1",
  "hours": [
    {
      "time": "2026-06-05T15:00:00Z",
      "overall_risk": 0.42,
      "rotor_risk": 0.18,
      "valley_wind_risk": 0.36,
      "foehn_risk": 0.04,
      "overdevelopment_risk": 0.22,
      "storm_growth_risk": 0.08,
      "reasons": [{"code": "increasing_gust_spread", "severity": "medium"}]
    }
  ]
}

```

HTTP	Code	Condition
403	tier_required	Detailed risk vectors require Pro.

GET /v1/thermals/sites/{site_id}/timeline

Field	Value
Owner service	thermal-service
Minimum tier	Pro
Summary	Return thermal and cloudbase forecast for site and surrounding XC zone.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=600

Response schema / example

```

{

```

```
"site_id": "site_vaga_001",
"hours": [
{
  "time": "2026-06-05T13:00:00Z",
  "thermal_strength_ms": 2.8,
  "thermal_height_m_msl": 1900,
  "cloudbase_m_msl": 1750,
  "cloudbase_confidence": 0.72,
  "trigger_quality": "moderate"
}
]
}
```

HTTP	Code	Condition
503	terrain_features_unavailable	Terrain/solar features not ready.

10. AI Briefing and Grounded Assistant APIs

AI endpoints are strictly grounded. The model receives only structured context assembled by the platform: forecast values, score outputs, risk vectors, site constraints, pilot profile, and rules. It may explain or summarize, but not invent weather, sites, airspace, legal rules, or safety claims.

POST /v1/briefings/site

Field	Value
Owner service	ai-briefing-service
Minimum tier	Pro+
Summary	Generate grounded site flight briefing.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_id": "site_vaga_001",
  "date": "2026-06-05",
  "pilot_profile_id": "prof_456",
  "question": "Can I fly there tomorrow and what is the best window?",
  "language": "en"
}
```

Response schema / example

```
{
  "briefing_id": "brf_123",
  "status": "ready",
  "answer": "Best window appears to be 11:30-14:30. Main risk is gust spread increasing after 15:00.",
  "decision": "GO_WITH_MONITORING",
  "must_show_disclaimer": true,
  "grounding": {
    "site_id": "site_vaga_001",
    "model_blend_id": "blend_nordic_v2_20260604_06",
    "flyability_score_ids": ["score_1", "score_2"],
    "risk_vector_ids": ["risk_1"]
  },
  "refusals": []
}
```

HTTP	Code	Condition
403	tier_required	AI briefing requires Pro+.
422	ungrounded_question	Question asks for unsupported data.

POST /v1/briefings/chat

Field	Value
Owner service	ai-briefing-service
Minimum tier	Pro+
Summary	Grounded conversational assistant for paragliding weather decisions.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "conversation_id": "conv_123",
  "message": "Compare Vaga and Site B for an intermediate pilot tomorrow.",
  "context": {"site_ids": ["site_vaga_001", "site_b_001"], "date": "2026-06-05"}
}
```

```
}
```

Response schema / example

```
{  
  "conversation_id": "conv_123",  
  "message_id": "msg_456",  
  "answer": "Vaga has a higher flyability score during 12:00-14:00, while Site B has lower gust spread but weaker thermal potential.",  
  "grounding_context_ids": ["ctx_1001", "ctx_1002"],  
  "limitations": ["No live station report available for Site B after 10:00Z."]  
}
```

HTTP	Code	Condition
422	insufficient_context	Assistant cannot answer without site/time context.

GET /v1/briefings/{briefing_id}/grounding

Field	Value
Owner service	ai-briefing-service
Minimum tier	Pro+
Summary	Return structured evidence used by an AI briefing.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=3600

Response schema / example

```
{  
  "briefing_id": "brf_123",  
  "inputs": {  
    "site": {"id": "site_vaga_001", "allowed_wind_sectors": [{"from_deg": 285, "to_deg": 45}]},  
    "forecast_hours": ["fc_1", "fc_2"],  
    "risk_vectors": ["risk_1"],  
    "scoring_version": "flyability-v4.2"  
  }  
}
```

HTTP	Code	Condition
404	briefing_not_found	No such briefing or no access.

11. XC Planning APIs

POST /v1/xc/plans

Field	Value
Owner service	xc-planner-service
Minimum tier	Pro+
Summary	Create XC route plan using weather, thermals, terrain, landings, airspace, and pilot profile.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "start_site_id": "site_vaga_001",
  "date": "2026-06-05",
  "target_distance_km": 50,
  "pilot_profile_id": "prof_456",
  "constraints": {
    "avoid_airspace": true,
    "prefer_safe_landings": true,
    "max_glide_ratio": 8.0,
    "latest_landing_time": "2026-06-05T18:00:00Z"
  }
}
```

Response schema / example

```
{
  "plan_id": "xcplan_123",
  "status": "ready",
  "summary": {
    "distance_km": 51.8,
    "estimated_duration_minutes": 135,
    "confidence": 0.68,
    "risk_level": "medium"
  },
  "route": {
    "type": "LineString",
    "coordinates": [[9.096,61.875],[9.30,61.92]]
  },
  "turnpoints": [{"lat": 61.89, "lon": 9.20, "reason": "thermal_trigger"}],
  "landing_options": [{"id": "landing_77", "quality": "good"}],
  "airspace_warnings": []
}
```

HTTP	Code	Condition
403	tier_required	XC planner requires Pro+.
422	no_safe_route	No route satisfies safety constraints.

GET /v1/xc/plans/{plan_id}

Field	Value
Owner service	xc-planner-service
Minimum tier	Pro+
Summary	Read generated XC route plan.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=3600

Response schema / example

```
{"plan_id": "xcplan_123", "status": "ready", "route": {}, "summary": {}}
```

HTTP	Code	Condition
404	plan_not_found	No such plan.

POST /v1/xc/plans/{plan_id}/export

Field	Value
Owner service	xc-planner-service
Minimum tier	Pro+
Summary	Export XC plan to GPX, KML, or app-native format.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"format": "gpx"}
```

Response schema / example

```
{"download_url": "https://files.example/xcplan_123.gpx", "expires_in_seconds": 3600}
```

HTTP	Code	Condition
400	unsupported_format	Format not supported.

12. Live Tracking, Telemetry, and WebSocket APIs

Tracking APIs support normal flight logs, live sharing, event tracking, and emergency scenarios. The mobile client batches telemetry when offline and submits with monotonic sequence numbers. The backend deduplicates by track_id + sequence.

POST /v1/tracks

Field	Value
Owner service	tracking-service
Minimum tier	Free
Summary	Start a flight track.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_id": "site_vaga_001",
  "visibility": "private",
  "device_id": "dev_123",
  "started_at": "2026-06-05T11:52:00Z"
}
```

Response schema / example

```
{
  "track_id": "trk_123",
  "upload_url": "/v1/tracks/trk_123/points",
  "ws_channel": "track.trk_123"
}
```

HTTP	Code	Condition
400	invalid_start_time	Start time too far in past/future.

POST /v1/tracks/{track_id}/points

Field	Value
Owner service	tracking-service
Minimum tier	Free
Summary	Upload batched telemetry points.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "points": [
    {
      "seq": 104,
      "time": "2026-06-05T12:01:03Z",
      "lat": 61.876,
      "lon": 9.102,
      "altitude_m": 1120,
      "ground_speed_kmh": 24.3,
      "vario_ms": 1.2,
      "battery_pct": 84
    }
  ]
}
```

Response schema / example

```
{"accepted": 1, "duplicates": 0, "last_seq": 104}
```

HTTP	Code	Condition
409	track_closed	Cannot append to closed track.
413	batch_too_large	Point batch exceeds configured limit.

POST /v1/tracks/{track_id}/finish

Field	Value
Owner service	tracking-service
Minimum tier	Free
Summary	Finish a track and trigger debrief pipeline.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"finished_at": "2026-06-05T14:20:00Z", "landing_site_id": "landing_77"}
```

Response schema / example

```
{"track_id": "trk_123", "status": "processing", "debrief_job_id": "job_888"}
```

HTTP	Code	Condition
409	already_finished	Track already finished.

WebSocket connection

```
GET /ws/v1?token=<jwt>&client_version=1.0.0
```

Client -> server subscribe:

```
{"op": "subscribe", "channels": ["site.site_vaga_001.flyability", "track.trk_123", "alerts.me"]}
```

Server event:

```
{"event": "flyability.updated", "channel": "site.site_vaga_001.flyability", "cursor": "0000002399", "payload": {"site_id": "site_vaga_001", "status": "MAYBE"}}
```

Client resume:

```
{"op": "resume", "channel": "alerts.me", "cursor": "0000002399"}
```

Channel	Publisher	Payload
site.{id}.flyability	decision-service	Changed GO/MAYBE/NO-GO status and top reason.
site.{id}.station	station-ingestion-service	Live station wind/gust/temperature.
track.{id}	tracking-service	Live telemetry point or track state.
event.{id}.leaderboard	event-service	Competition rank/task update.
alerts.me	notification-service	Personal safety and site alerts.
ops.incident	ops-service	Admin operational incidents.

13. Alerting and Notification APIs

POST /v1/alerts/rules

Field	Value
Owner service	notification-service
Minimum tier	Free/Pro
Summary	Create user alert rule for site/weather/flyability conditions.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "name": "Vaga good window",
  "site_id": "site_vaga_001",
  "enabled": true,
  "conditions": {
    "status_in": ["GO"],
    "min_cloudbase_m_agl": 1000,
    "max_gust_kmh": 28,
    "wind_direction_sector": {"from_deg": 290, "to_deg": 30}
  },
  "delivery": {"push": true, "email": false},
  "quiet_hours": {"from": "22:00", "to": "07:00", "timezone": "Europe/Oslo"}
}
```

Response schema / example

```
{"alert_rule_id": "alr_123", "status": "enabled"}
```

HTTP	Code	Condition
403	tier_required	User exceeded alert-rule limit for tier.
400	invalid_condition	Invalid alert rule condition.

GET /v1/alerts/rules

Field	Value
Owner service	notification-service
Minimum tier	Free
Summary	List own alert rules.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=60

Response schema / example

```
{"data": [{"id": "alr_123", "name": "Vaga good window", "enabled": true}]}
```

HTTP	Code	Condition
401	unauthorized	Missing token.

POST /v1/alerts/test

Field	Value
Owner service	notification-service
Minimum tier	Free
Summary	Send test push/email notification to current user.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"channel": "push", "device_id": "dev_123"}
```

Response schema / example

```
{"status": "queued"}
```

HTTP	Code	Condition
429	rate_limited	Too many tests.

14. Community Observation and Validation APIs

POST /v1/observations/pilot-reports

Field	Value
Owner service	observation-service
Minimum tier	Free
Summary	Submit pilot observation from launch or flight.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_id": "site_vaga_001",
  "observed_at": "2026-06-05T12:05:00Z",
  "location": {"lat": 61.875, "lon": 9.096},
  "wind_kmh": 18,
  "gust_kmh": 25,
  "wind_dir_deg": 310,
  "cloudbase_m_msl": 1700,
  "thermal_strength_ms": 2.5,
  "comment": "Good cycles, gusts building slowly.",
  "confidence": "measured_by_wind_meter"
}
```

Response schema / example

```
{"report_id": "obs_123", "quality_status": "pending_validation"}
```

HTTP	Code	Condition
409	duplicate_report	Similar report exists in time/location window.

POST /v1/observations/{report_id}/votes

Field	Value
Owner service	observation-service
Minimum tier	Free
Summary	Validate or dispute a pilot report.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"vote": "confirm", "reason": "Matches my measurement"}
```

Response schema / example

```
{"report_id": "obs_123", "validation_score": 0.78}
```

HTTP	Code	Condition
403	self_vote_not_allowed	Cannot vote on own report.

GET /v1/observations/sites/{site_id}/latest

Field	Value
Owner service	observation-service
Minimum tier	Free
Summary	Return latest validated observations for a site.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=60

Response schema / example

```
{"data": [{"report_id": "obs_123", "wind_kmh": 18, "gust_kmh": 25, "observed_at": "2026-06-05T12:05:00Z"}]}
```

HTTP	Code	Condition
404	site_not_found	Unknown site id.

15. Digital Twin and ML Correction APIs

Digital twin APIs are mostly internal/private in early releases. Public Pro+ outputs expose corrected forecasts and confidence, while raw feature vectors and model diagnostics remain internal/admin-only.

GET /v1/twins/sites/{site_id}/corrections

Field	Value
Owner service	digital-twin-service
Minimum tier	Pro+
Summary	Return site-specific corrected forecast deltas.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=600

Response schema / example

```
{
  "site_id": "site_vaga_001",
  "twin_version": "site-twin-v2.3",
  "corrections": [
    {
      "time": "2026-06-05T12:00:00Z",
      "surface_wind_delta_kmh": 3.2,
      "gust_delta_kmh": 1.8,
      "cloudbase_delta_m": -120,
      "confidence": 0.74,
      "training_sample_count": 428
    }
  ]
}
```

HTTP	Code	Condition
403	tier_required	Corrected forecasts require Pro+.
404	twin_not_ready	No trained twin for site.

POST /v1/internal/twins/training-runs

Field	Value
Owner service	digital-twin-service
Minimum tier	Internal
Summary	Start digital-twin training run.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_ids": ["site_vaga_001"],
  "from": "2024-01-01T00:00:00Z",
  "to": "2026-06-01T00:00:00Z",
  "model_family": "xgboost_regressor"
}
```

Response schema / example

```
{"training_run_id": "train_123", "status": "queued"}
```

HTTP	Code	Condition
403	role_required	Internal ML role required.

GET /v1/admin/twins/sites/{site_id}/quality

Field	Value
Owner service	digital-twin-service
Minimum tier	Admin

Summary	Inspect digital-twin performance and drift.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=300

Response schema / example

```
{
  "site_id": "site_vaga_001",
  "metrics": {
    "wind_mae_kmh_before": 5.8,
    "wind_mae_kmh_after": 3.9,
    "gust_mae_kmh_before": 7.2,
    "gust_mae_kmh_after": 5.4,
    "sample_count": 428
  },
  "drift": {"status": "normal", "last_checked": "2026-06-04T08:00:00Z"}
}
```

HTTP	Code	Condition
403	role_required	Admin or ML ops role required.

16. Airspace APIs

GET /v1/airspace/query

Field	Value
Owner service	airspace-service
Minimum tier	Free/Pro
Summary	Query airspace polygons intersecting bbox, route, or point/altitude.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=86400

Response schema / example

```
{
  "data": [
    {
      "id": "asp_123",
      "name": "Example CTR",
      "class": "D",
      "lower_limit": {"value": 0, "unit": "ft_agl"},
      "upper_limit": {"value": 2500, "unit": "ft_msl"},
      "geometry": {"type": "Polygon", "coordinates": []},
      "source": "openaip_import_20260604"
    }
  ]
}
```

HTTP	Code	Condition
400	invalid_geometry	Query geometry invalid.

POST /v1/airspace/check-route

Field	Value
Owner service	airspace-service
Minimum tier	Pro+
Summary	Check XC route or live track against airspace and altitude profile.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "route": {"type": "LineString", "coordinates": [[9.096,61.875],[9.30,61.92]]},
  "altitude_profile": [{"distance_km": 0, "altitude_m_msl": 950}, {"distance_km": 10, "altitude_m_msl": 1600}]
}
```

Response schema / example

```
{
  "violations": [],
  "warnings": [
    {"airspace_id": "asp_123", "distance_km": 18.2, "severity": "warning", "message": "Approaches controlled airspace margin."}
  ]
}
```

HTTP	Code	Condition
422	route_out_of_bounds	Route too large or unsupported.

17. Club, School, Event, and Rescue APIs

POST /v1/clubs

Field	Value
Owner service	organization-service
Minimum tier	Club
Summary	Create club organization.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"name": "Vaga Paragliding Club", "country": "NO"}
```

Response schema / example

```
{"club_id": "club_123", "role": "club_admin"}
```

HTTP	Code	Condition
403	tier_required	Club plan required.

POST /v1/clubs/{club_id}/sites

Field	Value
Owner service	organization-service
Minimum tier	Club
Summary	Associate managed site with club dashboard.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"site_id": "site_vaga_001", "management_note": "Local club validates reports."}
```

Response schema / example

```
{"club_id": "club_123", "site_id": "site_vaga_001", "status": "linked"}
```

HTTP	Code	Condition
403	organization_access_denied	User not club admin.

POST /v1/schools/{school_id}/training-windows

Field	Value
Owner service	school-service
Minimum tier	School
Summary	Generate conservative training windows for students.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_ids": ["site_vaga_001"],
  "date": "2026-06-05",
  "student_level": "first_high_flights",
  "instructor_count": 2
}
```

Response schema / example

```
{
  "windows": [
    {
      "site_id": "site_vaga_001",
      "from": "2026-06-05T10:00:00Z",
      "to": "2026-06-05T12:00:00Z",
      "status": "OK_FOR_TRAINING"
    }
  ]
}
```

HTTP	Code	Condition
403	tier_required	School plan required.

POST /v1/events/{event_id}/tasks

Field	Value
Owner service	event-service
Minimum tier	Event
Summary	Create competition task with weather and airspace validation.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "name": "Task 1",
  "turnpoints": [{"lat":61.875,"lon":9.096},{"lat":61.95,"lon":9.40}],
  "start_time": "2026-06-05T12:00:00Z",
  "goal_type": "line"
}
```

Response schema / example

```
{"task_id":"task_123","validation":{"weather":"marginal_after_16z","airspace":"ok"}}
```

HTTP	Code	Condition
422	invalid_task_geometry	Task geometry invalid.

POST /v1/rescue/search-area

Field	Value
Owner service	rescue-service
Minimum tier	Rescue
Summary	Estimate search area from last known telemetry and wind drift.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "track_id": "trk_123",
  "last_known_time": "2026-06-05T13:42:00Z",
  "search_start_time": "2026-06-05T15:10:00Z"
}
```

Response schema / example

```
{
  "search_area": {"type":"Polygon","coordinates": []},
  "confidence": 0.62,
  "assumptions": ["No telemetry after 13:42Z", "Wind from blend_nordic_v2_20260604_06"]
}
```

HTTP	Code	Condition
403	role_required	Rescue role required.
404	track_not_found	No access to requested track.

18. Billing, Subscriptions, and Entitlement APIs

GET /v1/billing/plans

Field	Value
Owner service	billing-service
Minimum tier	Anonymous
Summary	List public subscription plans and feature limits.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	public, max-age=3600

Response schema / example

```
{
  "data": [
    {"id": "free", "price_eur_month": 0, "features": {"forecast_horizon_hours": 72, "max_alert_rules": 3}},
    {"id": "pro", "price_eur_month": 4.99, "features": {"forecast_horizon_hours": 240, "max_alert_rules": 25}},
    {"id": "pro_plus", "price_eur_month": 9.99, "features": {"ai_briefing": true, "xc_planner": true}}
  ]
}
```

POST /v1/billing/checkout-session

Field	Value
Owner service	billing-service
Minimum tier	Free
Summary	Create payment provider checkout session.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{"plan_id": "pro_plus_yearly", "success_url": "https://app.example/success", "cancel_url": "https://app.example/cancel"}
```

Response schema / example

```
{"checkout_url": "https://billing.example/session/abc", "expires_at": "2026-06-04T11:15:00Z"}
```

HTTP	Code	Condition
400	invalid_plan	Unknown or unavailable plan.

POST /v1/billing/webhooks/payment-provider

Field	Value
Owner service	billing-service
Minimum tier	Internal/Public webhook
Summary	Receive signed payment events and update entitlements.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

Raw signed provider payload. Must verify signature before processing.

Response schema / example

```
{"status": "accepted"}
```

HTTP	Code	Condition
400	invalid_signature	Webhook signature invalid.
409	duplicate_event	Webhook event already processed.

19. Admin, Ops, and Data-Quality APIs

GET /v1/ops/health

Field	Value
Owner service	ops-service
Minimum tier	Internal/Admin
Summary	Return aggregate platform health.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Response schema / example

```
{
  "status":"degraded",
  "services":[
    {"name":"forecast-processing-service","status":"ok"},
    {"name":"station-ingestion-service","status":"degraded","reason":"provider timeout"}
  ]
}
```

GET /v1/admin/data-quality/sites/{site_id}

Field	Value
Owner service	ops-service
Minimum tier	Admin
Summary	Inspect data quality for a site.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	private, max-age=60

Response schema / example

```
{
  "site_id":"site_vaga_001",
  "forecast_freshness_seconds":420,
  "station_freshness_seconds":90,
  "observation_count_24h":6,
  "known_gaps":["No webcam feed"],
  "decision_readiness":"ready"
}
```

HTTP	Code	Condition
403	role_required	Admin role required.

POST /v1/admin/site-safety-overrides

Field	Value
Owner service	ops-service
Minimum tier	Admin
Summary	Apply temporary site closure or safety override.
Idempotency	Required for POST/PUT/PATCH unless endpoint is read-only
Cache policy	no-store

Request schema / example

```
{
  "site_id":"site_vaga_001",
  "status":"CLOSED",
  "reason":"Local authority closure due to rescue operation",
  "from":"2026-06-05T09:00:00Z",
  "to":"2026-06-05T18:00:00Z"
}
```

Response schema / example

```
{"override_id":"ovr_123","status":"active"}
```

HTTP	Code	Condition
409	overlap_conflict	Conflicts with existing override.

20. OpenAPI File Structure and Implementation Backlog

The implementation repository should store API contracts independently from service code so backend, frontend, QA, and documentation can share the same source of truth.

/api-contracts

/openapi

platform.v1.yaml

auth.v1.yaml

sites.v1.yaml

weather.v1.yaml

flyability.v1.yaml

tracking.v1.yaml

billing.v1.yaml

/schemas

geojson.schema.json

error.schema.json

forecast-hour.schema.json

flyability-score.schema.json

/examples

site-detail.json

site-forecast-hourly.json

flyability-timeline.json

websocket-events.json

Backlog item	Priority	Definition of done
Generate OpenAPI 3.1 YAML from this document	P0	All endpoint paths, methods, auth, responses, and error codes represented.
Build API gateway policy bundle	P0	Rate limits, entitlement checks, CORS, JWT validation, request-id propagation.
Create contract tests	P0	Consumer-driven tests for mobile and web clients using example payloads.
Add mock server	P1	Frontend can develop against deterministic API mocks.
Implement SDK generation	P1	Typescript and Kotlin/Swift models generated from OpenAPI schemas.
Add websocket load tests	P1	Sustain target concurrent connections and reconnect/resume behavior.
Add safety traceability tests	P0	Every decision response must include source run IDs and scoring version.

21. References and External Integration Baselines

These references informed the API design and should be verified during implementation procurement and licensing review:

Source	URL	Use in platform
Windy Map Forecast API	https://api.windy.com/map-forecast/docs	Weather map visualizations, layers, particles, isolines, and map interaction baseline.
Open-Meteo Forecast API	https://open-meteo.com/en/docs	Multi-model forecast API and normalized hourly time-series baseline for MVP ingestion.
MET Norway Weather API	https://api.met.no/	Norwegian weather API access, licensing, and locationforecast/nowcast integration.
MET Norway Locationforecast	https://api.met.no/weatherapi/locationforecast/2.0/documentation	Coordinate forecast service characteristics and limitations.
OpenAIP API Documentation	https://docs.openaip.net/	Aeronautical data and airspace integration baseline.
PostGIS Documentation	https://postgis.net/docs/	Spatial database behavior and GIS query assumptions.
TimescaleDB Documentation	https://docs.timescale.com/	Timeseries hypertables, retention, and continuous aggregates.
NOAA NOMADS	https://nomads.ncep.noaa.gov/	GFS model access baseline.
ECMWF Open Data	https://www.ecmwf.int/en/forecasts/datasets/open-data	ECMWF forecast dataset baseline.